

WEEK – 12

DURATION: 29 March 2026 – 03 April 2026

DAY 2: 30 March 2026

1. TASKS WORKED ON

- ***Controlled and Uncontrolled Components in React:*** Learned and implemented controlled and uncontrolled components in React for handling form inputs, validation, and submission. Also conducted a peer learning session with Jagjot and Harsh to explain their differences, use cases, and implementation approaches.
- **Netsquare Softwares Website – Content Gathering:** Worked on content collection, structuring, and updates for the Netsquare Softwares website by analyzing the existing website and aligning content with the new design. Updated sections included Services, Paid Ads Promotion, Consultation, Career, and Work pages.

2. LEARNING AND KNOWLEDGE GAINED

Today, I learned how to manage and validate form inputs in React using both controlled and uncontrolled component approaches. I improved my understanding of state-based and ref-based form handling techniques along with their practical applications in frontend development.

3. PLAN FOR NEXT WORKING DAY

- Learn additional React concepts and advanced state management techniques.
- Continue working on live projects and frontend development tasks.

DAY 3: 01 April 2026

1. TASKS WORKED ON

- **Study of Tracking Truck Website:** Completed the frontend and backend setup for the Tracking Truck project, including environment configuration, database connectivity, and API integrations. Also studied the overall project workflow to understand authentication, dashboard functionality, module structure, and truck tracking operations.
- **Aurevia Website Project Study:** Analyzed the reference and existing Aurevia websites to understand design, animations, navigation, and areas for improvement. Also completed the initial project setup using Next.js, Tailwind CSS, GSAP, and other required dependencies while planning the project structure and workflow.
- **Advanced State Management – Context API to Redux:** Learned and implemented the React Context API for global state management by building a theme switcher and managing application state. Also explored advanced state handling using Context API with `useReducer()` for authentication and user preference management.

2. LEARNING AND KNOWLEDGE GAINED

Today, I improved my understanding of large-scale project workflows including frontend setup, backend configuration, and application architecture.

3. PLAN FOR NEXT WORKING DAY

- Learn additional advanced React concepts.
- Continue development work on the Aurevia website.
- Explore more animation techniques and UI implementation strategies.

DAY 4: 02 April 2026

1. TASKS WORKED ON

- **AWS S3 Concepts & Cloud Storage Understanding:** Learned the fundamentals of AWS S3 and cloud storage, including file storage, retrieval, security configurations, and bucket management. Also explored file upload workflows, scalability benefits, and different upload approaches such as server-side uploads and Presigned URLs
- **Aurevia Website Development:** Developed key UI components for the project, including an animated preloader, a responsive navbar, and an interactive Hero section. Used GSAP and ScrollTrigger to implement smooth animations, scrolling effects, and an enhanced user experience.

2. LEARNING AND KNOWLEDGE GAINED

Today, I gained practical knowledge of AWS S3 cloud storage, secure file handling, IAM configurations, and file upload strategies using Presigned URLs. I also enhanced my understanding of GSAP animations for creating interactive user interfaces and strengthened my knowledge of Redux-style state management patterns in React applications.

3. PLAN FOR NEXT WORKING DAY

- Learn additional React concepts and advanced frontend techniques.
- Continue development work on the Aurevia website.
- Explore more AWS services and cloud integration concepts.

DAY 5: 03 April 2026

1. TASKS WORKED ON

- **Full-Stack AWS S3 Integration;** Developed a complete file upload system by integrating a React frontend, Node.js backend, AWS S3 storage, and CloudFront CDN. Also implemented database storage for file metadata and applied security and optimization practices to improve scalability and performance.
- **React Routing & State Management Mini Project:** Implemented protected routes for authentication-based access control and developed dynamic components using URL-based state management. Also created blog and catalog simulations to demonstrate dynamic content rendering and filtering functionality.

2. LEARNING AND KNOWLEDGE GAINED

Today, I gained hands-on experience in building a full-stack application integrated with AWS S3 cloud storage and improved my understanding of secure file management, protected routing, dynamic navigation, and scalable application architecture.

3. PLAN FOR NEXT WORKING DAY

- Learn new advanced concepts related to React.
- Continue working on the “mb-usa-trucking” website and related frontend features.

GitHub Link:

<https://github.com/Gursimran-12/MERN-Stack-Internship/tree/main/Tasks-week-12>